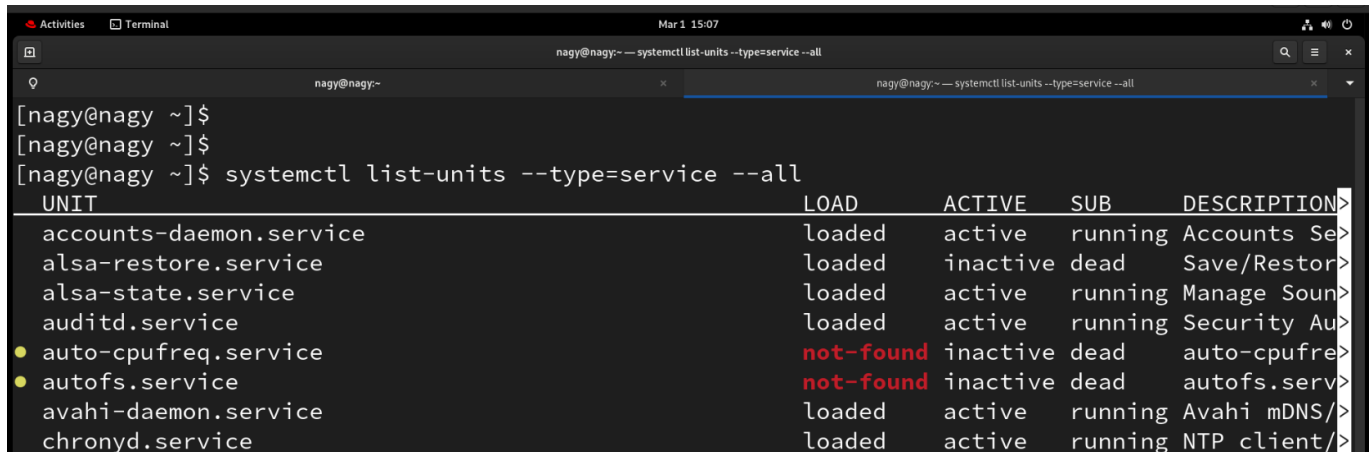


# SystemAdminII : Lab 1

## 1. Use systemctl to view the status of all the system services.

```
systemctl list-units --type=service --all
```

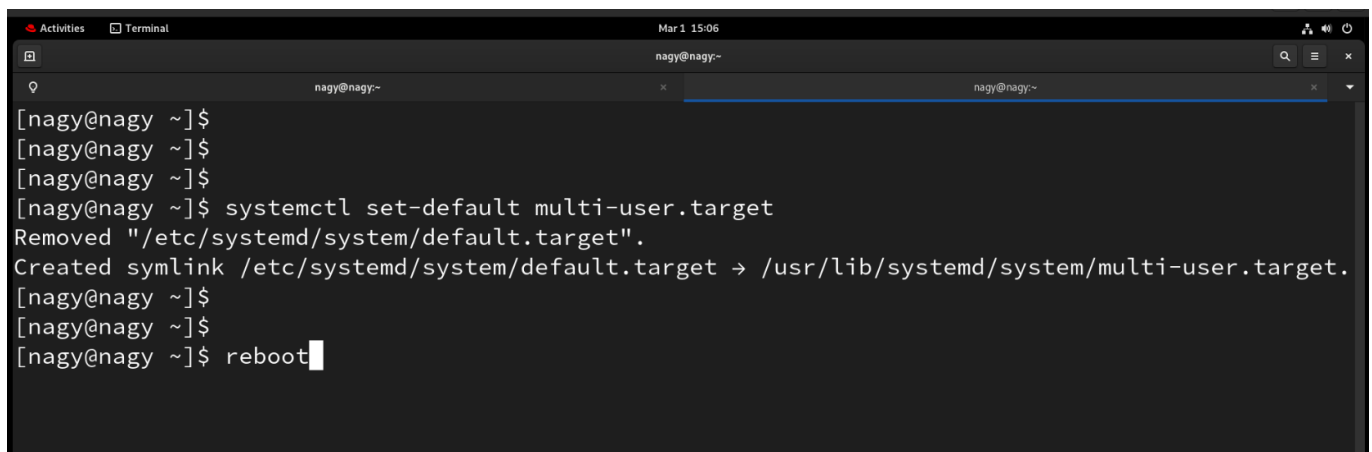


A terminal window showing the command `systemctl list-units --type=service --all` and its output. The output is a table with columns: UNIT, LOAD, ACTIVE, SUB, and DESCRIPTION. The units listed are accounts-daemon.service, alsa-restore.service, alsa-state.service, auditd.service, auto-cpufreq.service, autofs.service, avahi-daemon.service, and chronyd.service. The status for auto-cpufreq.service and autofs.service is 'not-found'.

| UNIT                    | LOAD      | ACTIVE   | SUB     | DESCRIPTION  |
|-------------------------|-----------|----------|---------|--------------|
| accounts-daemon.service | loaded    | active   | running | Accounts Se  |
| alsa-restore.service    | loaded    | inactive | dead    | Save/Restor  |
| alsa-state.service      | loaded    | active   | running | Manage Soun  |
| auditd.service          | loaded    | active   | running | Security Au  |
| auto-cpufreq.service    | not-found | inactive | dead    | auto-cpufre  |
| autofs.service          | not-found | inactive | dead    | autofs.serv  |
| avahi-daemon.service    | loaded    | active   | running | Avahi mDNS/> |
| chronyd.service         | loaded    | active   | running | NTP client/> |

## 2. Change the default run level back to multi-user.target and reboot.

```
systemctl set-default multi-user.target  
reboot
```

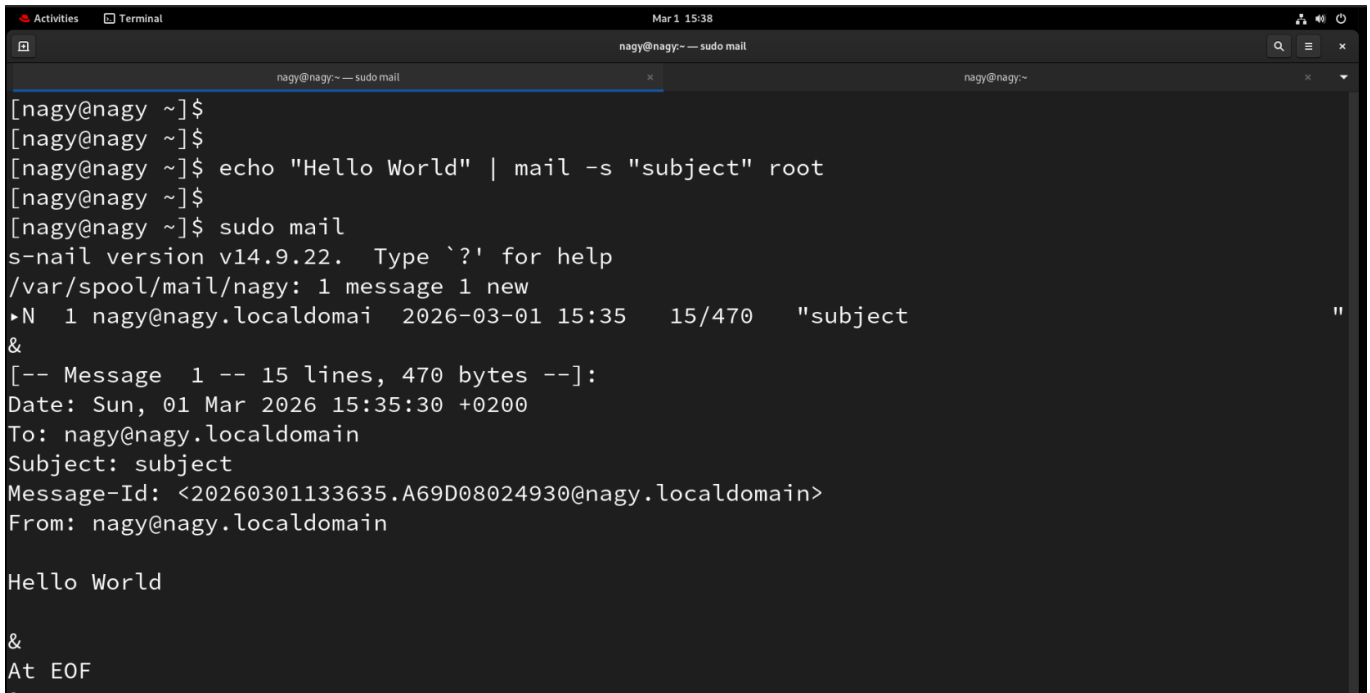


A terminal window showing the commands `systemctl set-default multi-user.target` and `reboot`. The output of the first command shows that the default target has been changed from `graphical.target` to `multi-user.target`.

```
[nagy@nagy ~]$ systemctl set-default multi-user.target  
Removed "/etc/systemd/system/default.target".  
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/multi-user.target.  
[nagy@nagy ~]$  
[nagy@nagy ~]$  
[nagy@nagy ~]$ reboot
```

## 3. Send mail to the root user.

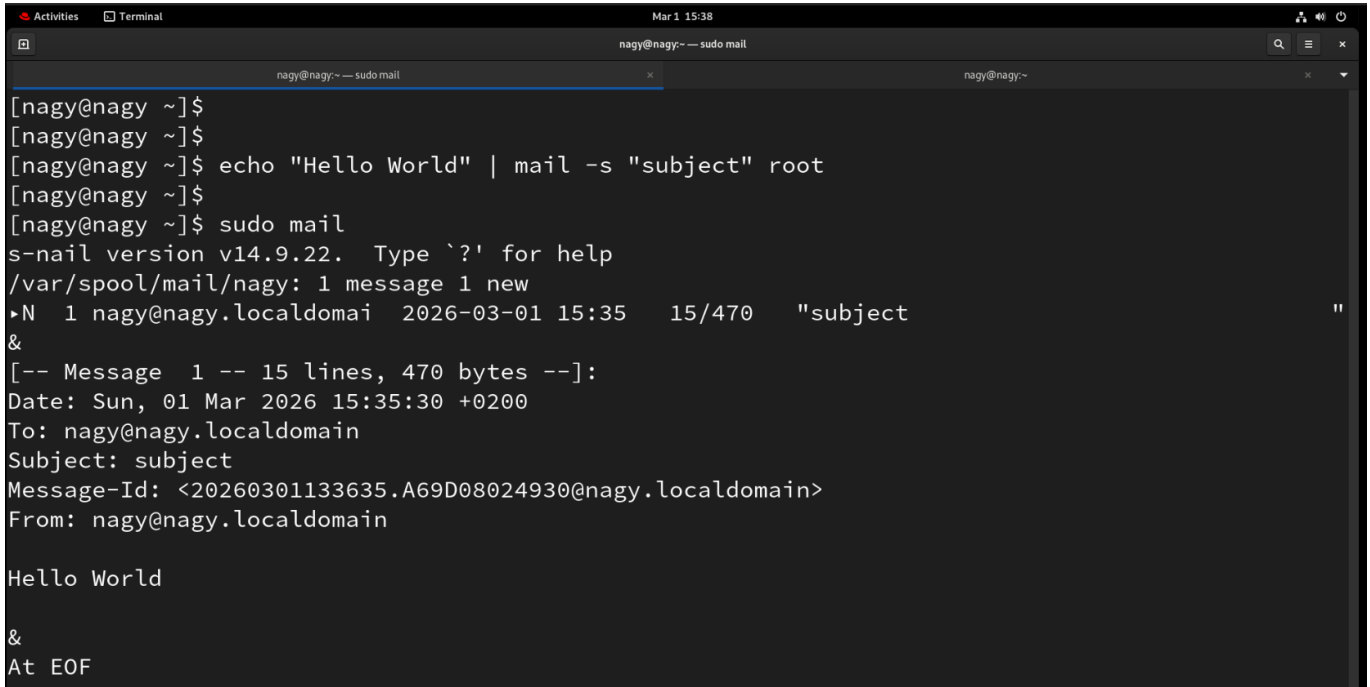
```
echo "Hello world" | mail -s "Subject" root
```



```
[nagy@nagy ~]$  
[nagy@nagy ~]$  
[nagy@nagy ~]$ echo "Hello World" | mail -s "subject" root  
[nagy@nagy ~]$  
[nagy@nagy ~]$ sudo mail  
s-nail version v14.9.22. Type '?' for help  
/var/spool/mail/nagy: 1 message 1 new  
*N 1 nagy@nagy.localdomai 2026-03-01 15:35 15/470 "subject"  
&  
[-- Message 1 -- 15 lines, 470 bytes --]:  
Date: Sun, 01 Mar 2026 15:35:30 +0200  
To: nagy@nagy.localdomain  
Subject: subject  
Message-Id: <20260301133635.A69D08024930@nagy.localdomain>  
From: nagy@nagy.localdomain  
  
Hello World  
  
&  
At EOF
```

#### 4. Verify that you have received this mail.

```
mail
```



```
[nagy@nagy ~]$  
[nagy@nagy ~]$  
[nagy@nagy ~]$ echo "Hello World" | mail -s "subject" root  
[nagy@nagy ~]$  
[nagy@nagy ~]$ sudo mail  
s-nail version v14.9.22. Type '?' for help  
/var/spool/mail/nagy: 1 message 1 new  
*N 1 nagy@nagy.localdomai 2026-03-01 15:35 15/470 "subject"  
&  
[-- Message 1 -- 15 lines, 470 bytes --]:  
Date: Sun, 01 Mar 2026 15:35:30 +0200  
To: nagy@nagy.localdomain  
Subject: subject  
Message-Id: <20260301133635.A69D08024930@nagy.localdomain>  
From: nagy@nagy.localdomain  
  
Hello World  
  
&  
At EOF
```

#### 5. Use systemctl utility to stop postfix service.

```
systemctl stop postfix  
systemctl status postfix
```

```
Activities Terminal Mar 1 15:39
nagy@nagy:~ — sudo systemctl status postfix

[nagy@nagy ~]$ sudo systemctl stop postfix
[sudo] password for nagy:
[nagy@nagy ~]$
[nagy@nagy ~]$ sudo systemctl status postfix
o postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; preset:
   Active: inactive (dead) since Sun 2026-03-01 15:39:24 EET; 4s ago
   Duration: 2min 48.495s
   Process: 60578 ExecStop=/usr/sbin/postfix stop (code=exited, status=0/SUCC
   Main PID: 59882 (code=killed, signal=TERM)
   CPU: 933ms

Mar 01 15:36:36 nagy postfix/qmgr[59884]: A5B41802492E: removed
Mar 01 15:37:44 nagy postfix/pickup[59883]: 4156A802492E: uid=1000 from=<nagy>
Mar 01 15:37:44 nagy postfix/cleanup[59887]: 4156A802492E: message-id=<2026030
Mar 01 15:37:44 nagy postfix/qmgr[59884]: 4156A802492E: from=<nagy@nagy.locald
Mar 01 15:37:44 nagy postfix/local[59893]: 4156A802492E: to=<root@nagy.locald
Mar 01 15:37:44 nagy postfix/qmgr[59884]: 4156A802492E: removed
Mar 01 15:39:24 nagy systemd[1]: Stopping Postfix Mail Transport Agent...
Mar 01 15:39:24 nagy postfix/master[59882]: terminating on signal 15
Mar 01 15:39:24 nagy systemd[1]: postfix.service: Deactivated successfully.
Mar 01 15:39:24 nagy systemd[1]: Stopped Postfix Mail Transport Agent.
lines 1-18/18 (END)...skipping...
```

## 6. Send mail again to the root user.

```
echo "Second test mail" | mail -s "Postfix Stopped Test" root
```

Mail will be queued because postfix is stopped.

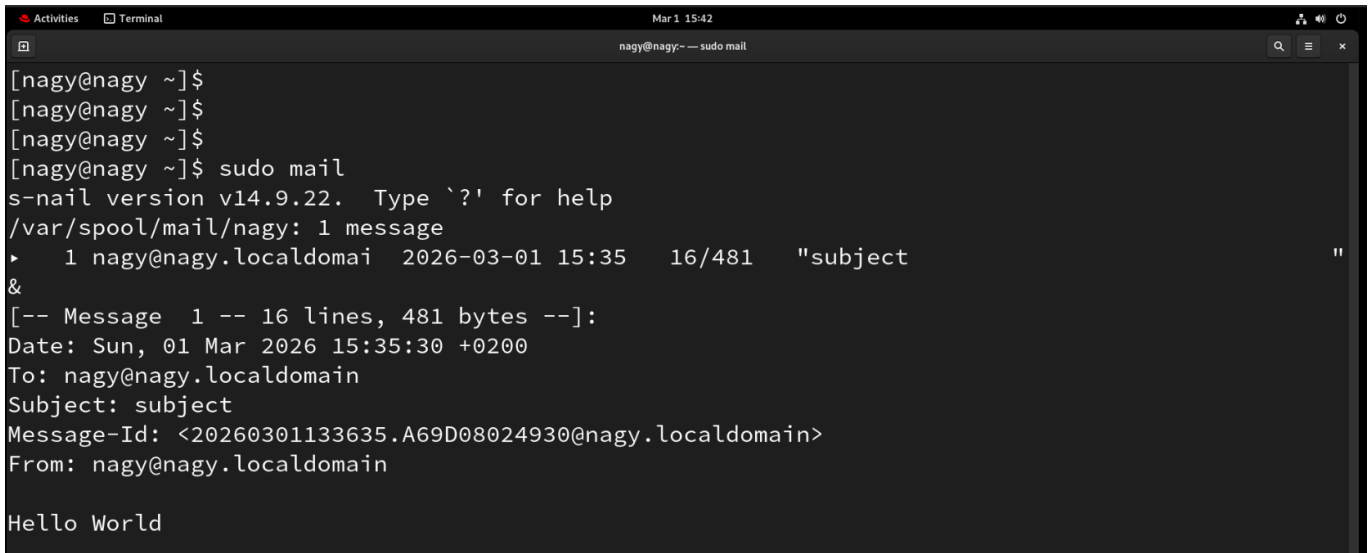
```
Activities Terminal Mar 1 15:42
nagy@nagy:~

~
[nagy@nagy ~]$
[nagy@nagy ~]$
[nagy@nagy ~]$
[nagy@nagy ~]$ echo "Second mail" | mail -s "subject" root
[nagy@nagy ~]$
[nagy@nagy ~]$
```

## 7. Verify that you have received this mail.

```
mail
```

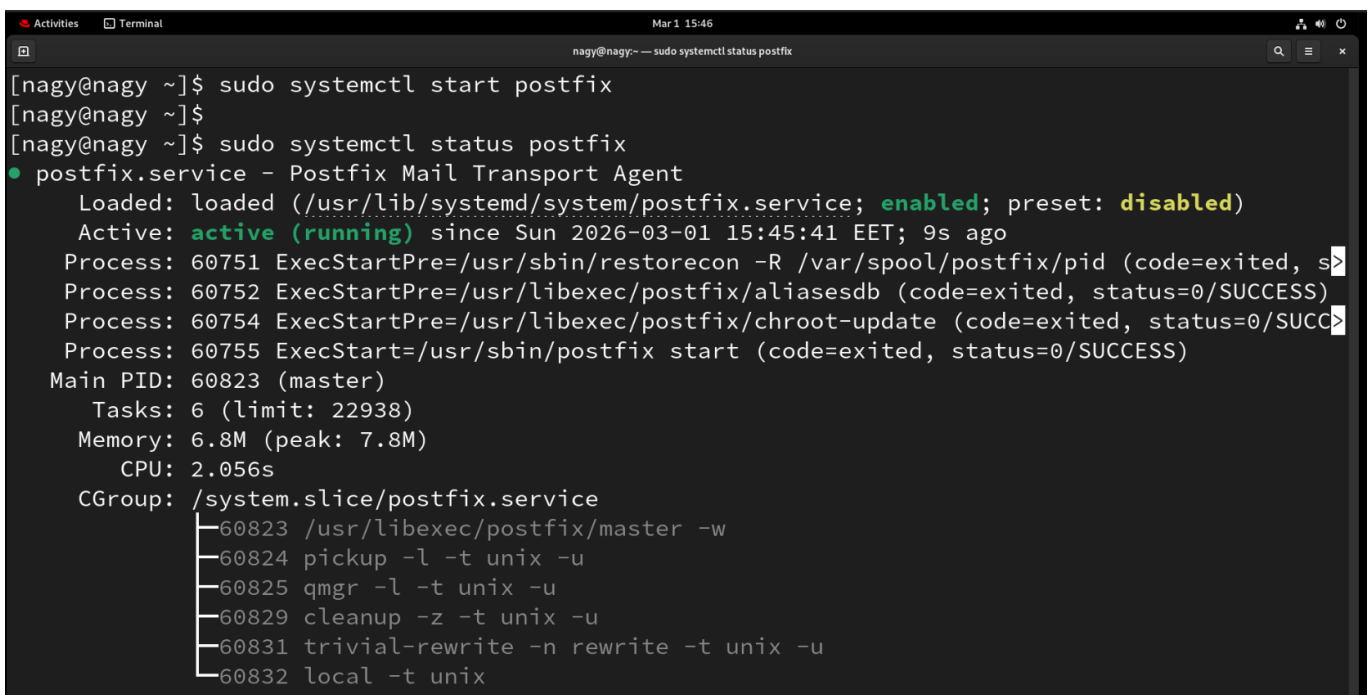
The new mail will not appear until postfix is started.



```
[nagy@nagy ~]$  
[nagy@nagy ~]$  
[nagy@nagy ~]$  
[nagy@nagy ~]$ sudo mail  
s-nail version v14.9.22. Type '?' for help  
/var/spool/mail/nagy: 1 message  
▶ 1 nagy@nagy.localdomain 2026-03-01 15:35 16/481 "subject"  
&  
[-- Message 1 -- 16 lines, 481 bytes --]:  
Date: Sun, 01 Mar 2026 15:35:30 +0200  
To: nagy@nagy.localdomain  
Subject: subject  
Message-Id: <20260301133635.A69D08024930@nagy.localdomain>  
From: nagy@nagy.localdomain  
  
Hello World
```

## 8. Use systemctl utility to start postfix service.

```
systemctl start postfix  
systemctl status postfix
```

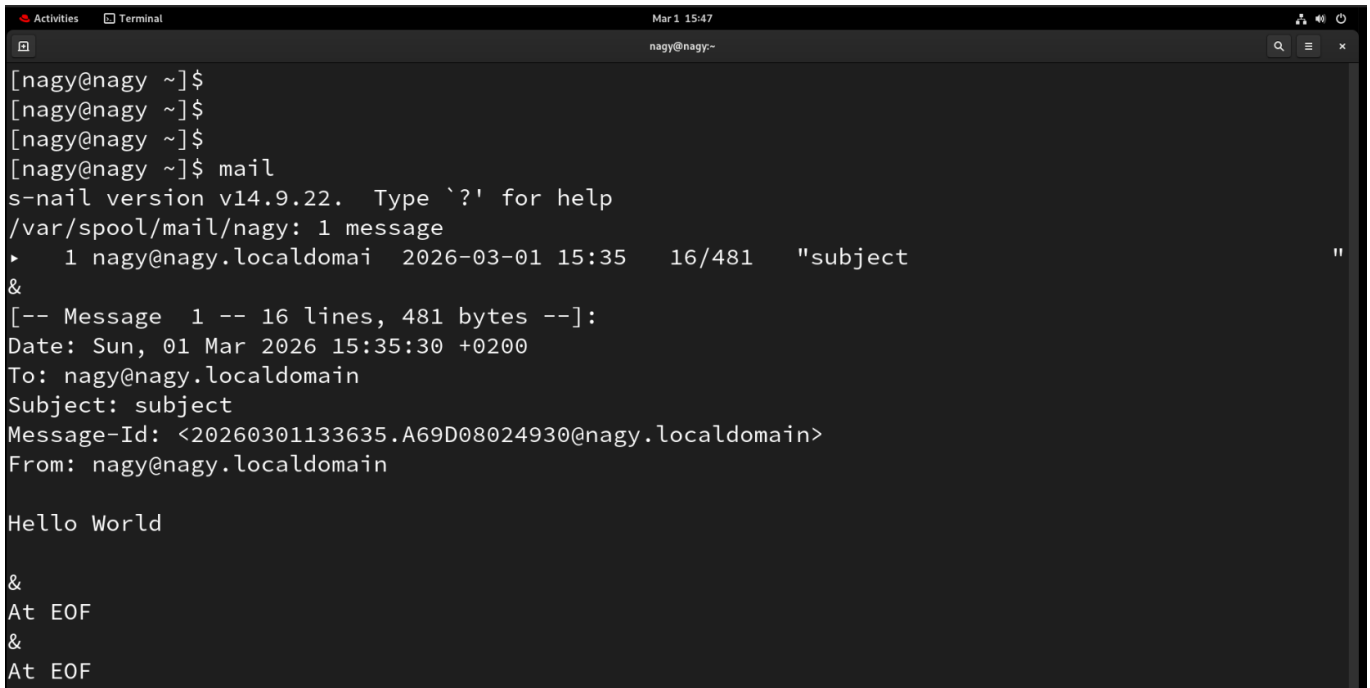


```
[nagy@nagy ~]$ sudo systemctl start postfix  
[nagy@nagy ~]$  
[nagy@nagy ~]$ sudo systemctl status postfix  
● postfix.service - Postfix Mail Transport Agent  
   Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; preset: disabled)  
   Active: active (running) since Sun 2026-03-01 15:45:41 EET; 9s ago  
     Process: 60751 ExecStartPre=/usr/sbin/restorecon -R /var/spool/postfix/pid (code=exited, status=0/SUCCESS)  
     Process: 60752 ExecStartPre=/usr/libexec/postfix/aliasesdb (code=exited, status=0/SUCCESS)  
     Process: 60754 ExecStartPre=/usr/libexec/postfix/chroot-update (code=exited, status=0/SUCCESS)  
     Process: 60755 ExecStart=/usr/sbin/postfix start (code=exited, status=0/SUCCESS)  
 Main PID: 60823 (master)  
    Tasks: 6 (limit: 22938)  
  Memory: 6.8M (peak: 7.8M)  
     CPU: 2.056s  
   CGroup: /system.slice/postfix.service  
           └─60823 /usr/libexec/postfix/master -w  
             └─60824 pickup -l -t unix -u  
               └─60825 qmgr -l -t unix -u  
                 └─60829 cleanup -z -t unix -u  
                   └─60831 trivial-rewrite -n rewrite -t unix -u  
                     └─60832 local -t unix
```

## 9. Verify that you have received this mail.

```
mail
```

The queued mail should now be delivered.

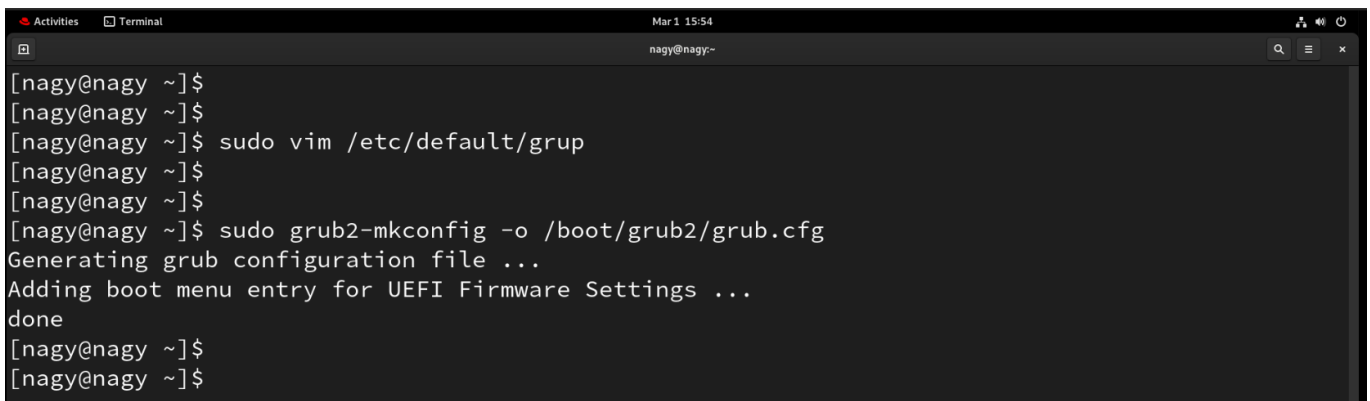
A terminal window titled 'Terminal' with a dark background. The prompt is 'nagy@nagy-'. The user enters 'mail'. The output shows 's-nail version v14.9.22. Type `?' for help' and '/var/spool/mail/nagy: 1 message'. It then lists a message from 'nagy@nagy.localdomain' with subject 'subject' and body 'Hello World'. The user enters '&' and 'At EOF' twice.

```
[nagy@nagy ~]$  
[nagy@nagy ~]$  
[nagy@nagy ~]$  
[nagy@nagy ~]$ mail  
s-nail version v14.9.22. Type `?' for help  
/var/spool/mail/nagy: 1 message  
  1 nagy@nagy.localdomain 2026-03-01 15:35 16/481 "subject"  
&  
[-- Message 1 -- 16 lines, 481 bytes --]:  
Date: Sun, 01 Mar 2026 15:35:30 +0200  
To: nagy@nagy.localdomain  
Subject: subject  
Message-Id: <20260301133635.A69D08024930@nagy.localdomain>  
From: nagy@nagy.localdomain  
  
Hello World  
  
&  
At EOF  
&  
At EOF
```

---

## 10. Edit GRUB2 configuration and change timeout to 20 seconds.

```
vim /etc/default/grub  
sudo grub2-mkconfig -o /boot/grub2/grub.cfg
```

A terminal window titled 'Terminal' with a dark background. The prompt is 'nagy@nagy-'. The user enters 'sudo vim /etc/default/grub' and 'sudo grub2-mkconfig -o /boot/grub2/grub.cfg'. The output shows 'Generating grub configuration file ...' and 'Adding boot menu entry for UEFI Firmware Settings ... done'.

```
[nagy@nagy ~]$  
[nagy@nagy ~]$  
[nagy@nagy ~]$ sudo vim /etc/default/grub  
[nagy@nagy ~]$  
[nagy@nagy ~]$  
[nagy@nagy ~]$ sudo grub2-mkconfig -o /boot/grub2/grub.cfg  
Generating grub configuration file ...  
Adding boot menu entry for UEFI Firmware Settings ...  
done  
[nagy@nagy ~]$  
[nagy@nagy ~]$
```

---

## 11. Change default operating system in GRUB2.

```
vim /etc/default/grub  
sudo grub2-mkconfig -o /boot/grub2/grub.cfg
```

```
Activities Terminal Mar 1 15:56
nagy@nagy~$
nagy@nagy~$
nagy@nagy~$ sudo vim /etc/default/grub
nagy@nagy~$
nagy@nagy~$ sudo grub2-mkconfig -o /boot/grub2/grub.cfg
Generating grub configuration file ...
Adding boot menu entry for UEFI Firmware Settings ...
done
nagy@nagy~$
nagy@nagy~$
nagy@nagy~$
```

---

12. Monitor memory every 10 minutes between 8:00 AM and 5:00 PM.

```
crontab -e
```

```
*/10 8-17 * * * free -m >> /var/log/memory.log
```

---

13. Check cron job mail as root.

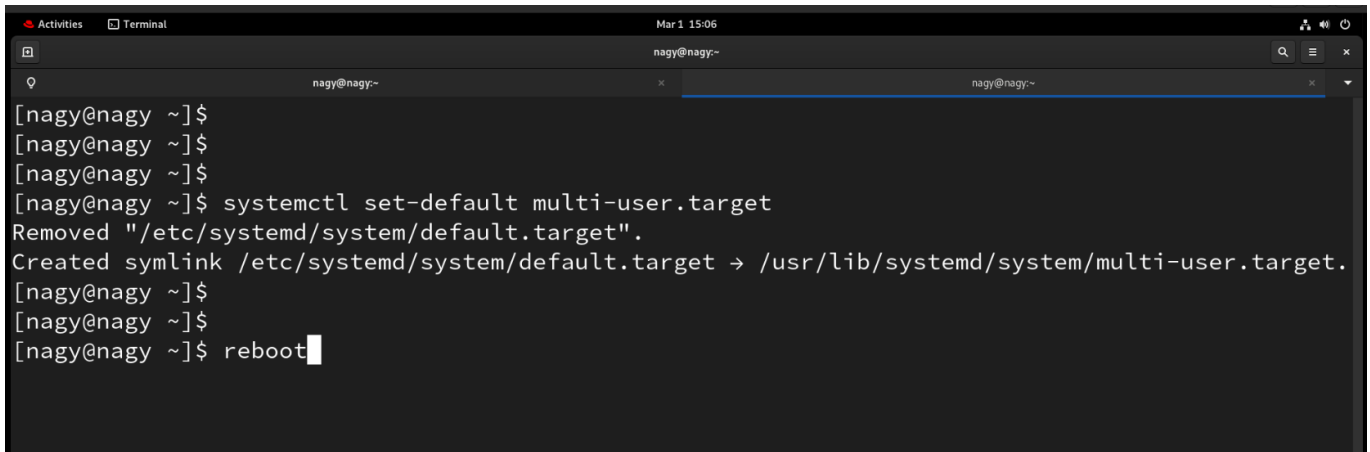
```
mail
```

---

14. Send cron output to another user (manager).

```
crontab -e
```

```
*/10 8-17 * * * free -m | mail -s "Memory Report" manager
```

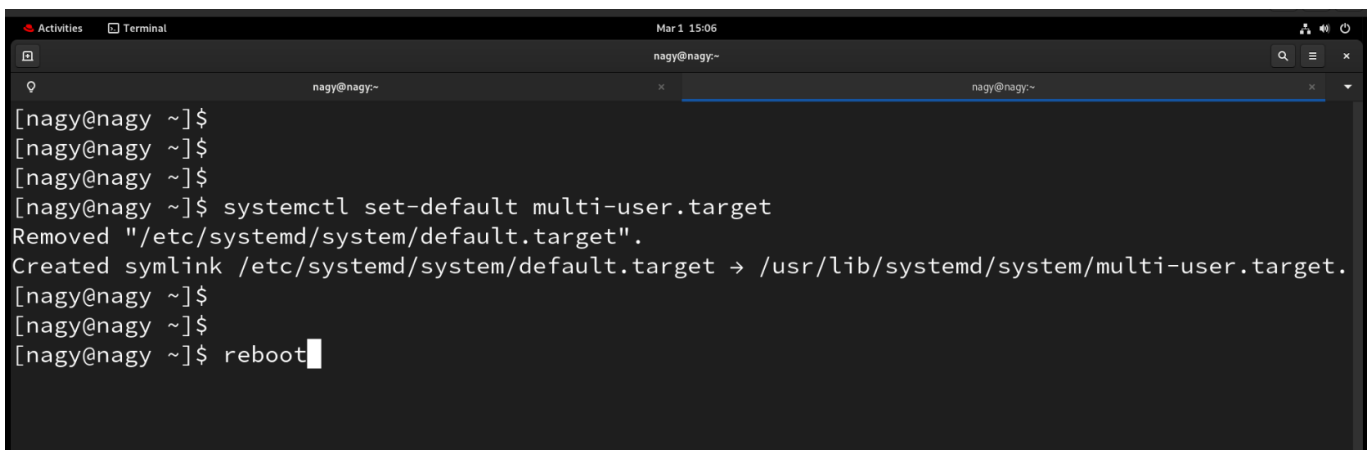
A terminal window with a dark background. The prompt is [nagy@nagy ~]\$. The user enters 'systemctl set-default multi-user.target'. The output shows the removal of the old default target and the creation of a symlink to the new one. The user then enters 'reboot' and the prompt returns.

```
[nagy@nagy ~]$  
[nagy@nagy ~]$  
[nagy@nagy ~]$  
[nagy@nagy ~]$ systemctl set-default multi-user.target  
Removed "/etc/systemd/system/default.target".  
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/multi-user.target.  
[nagy@nagy ~]$  
[nagy@nagy ~]$  
[nagy@nagy ~]$ reboot
```

---

## 15. Check cron mail as manager user.

```
su - manager  
mail
```

A terminal window with a dark background. The prompt is [nagy@nagy ~]\$. The user enters 'systemctl set-default multi-user.target'. The output shows the removal of the old default target and the creation of a symlink to the new one. The user then enters 'reboot' and the prompt returns.

```
[nagy@nagy ~]$  
[nagy@nagy ~]$  
[nagy@nagy ~]$  
[nagy@nagy ~]$ systemctl set-default multi-user.target  
Removed "/etc/systemd/system/default.target".  
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/multi-user.target.  
[nagy@nagy ~]$  
[nagy@nagy ~]$  
[nagy@nagy ~]$ reboot
```